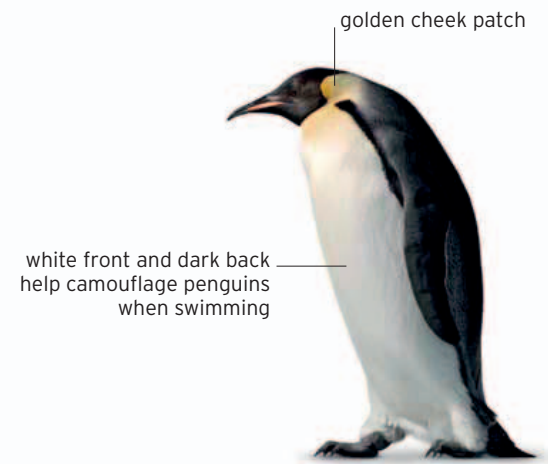


Emperor penguin

Aptenodytes forsteri



Adapted to some of the most challenging conditions on Earth, the emperor penguin is the only bird that breeds during the severe Antarctic winter. It is the largest of the penguins, but has the same upright pose, short legs, waddling walk, and stiff wings held like arms at its sides, as well as an incurably inquisitive nature.

Built for survival

When moving from ice-cold water into the warmth of the sun, penguins may wave their wings to dissipate heat and they also pant to keep cool. In cold conditions, emperor penguins tilt back onto their heels and tails, avoiding contact between the ice and their feet. The dark plumage absorbs heat from the sun and beneath the dense feathers is a trapped layer of insulating warm air. Beneath this they have a thick layer of fat—creating the familiar rotund, streamlined shape—that is indispensable for survival in temperatures as low as -76°F (-60°C).

Emperor penguins live in a narrow band of ice floes and frozen bays surrounding the Antarctic continent. The rookeries get further from the open sea as it freezes, forcing these flightless birds to undertake annual treks

to establish their colonies in March or April, and then make repeated and longer marches—up to 37 miles (60 km) each way—to bring food back to their chick.

Teamwork

The female lays a single egg, which she transfers to the male before heading to the open ocean on a feeding trip that lasts about two months. The male incubates the egg in his brood pouch, a fold of featherless skin just above his feet. Emperor penguin colonies may be several thousand strong, with the incubating males huddling together over large open areas. They stand almost motionless for days in low temperatures and raging blizzards. When the female returns, care of the newly hatched chick transfers to her and the near-starving male goes off to feed.

After 45 days, when it is well grown and covered in down, the chick joins a nursery, but it is still fed by its parents. The nursery breaks up after five months, when the parents abandon their chick and leave for the open sea. The chicks eventually follow and, having acquired adult plumage, take to the water.

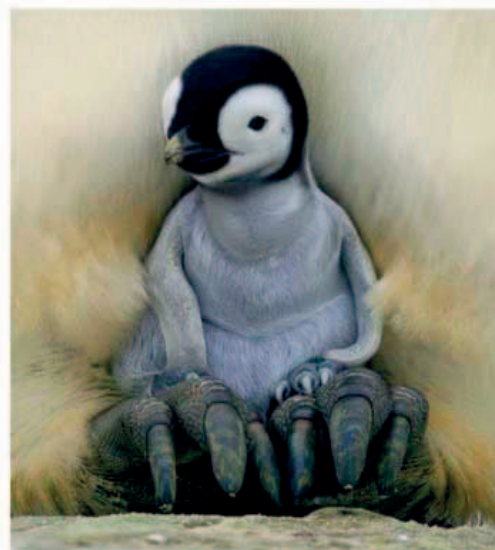
Emperor penguins **dive deeper than any other bird** to find food

◁ BUBBLE POWER

Penguin feathers release a stream of air bubbles when the birds head back to the surface from a deep dive. This increases their speed of ascent sufficiently for them to clear the water and land safely on the ice.

▷ KEEPING WARM

The solitary emperor penguin chick is hatched almost naked. It is kept warm by a parent until it is covered with down and big enough to join a nursery.



- ↔ 4 ft (1.1m)
- ⚖ 66–88 lb (30–40 kg)
- ⊗ Near threatened
- 🐠 Krill, fish, squid
- 🏠



Circumpolar around Antarctica